RAJALAKSHMI ENGINEERING COLLEGE

B.E. COMPUTER SCIENCE AND ENGINEERING | CYBER SECURITY

CYBER AND DIGITAL FORENSICS

CR23532 • CASE STUDY ASSIGNMENTS – II

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An overview of the five case studies

Part	Unique scenario title
2(1)	After-Hours Copy of an Examination Moderation Archive
2(2)	Shared Helpdesk PC and a Disguised Maintenance Executable
2(3)	Seizure of a Robotics Sponsorship Laptop and USB Drive
2(4)	Deleted Research Grant Ledger Recovered from Shadow Copies
2(5)	Compromised Campus Poster-Portal Account Across Five Sources

This submission covers all 38 questions across five separate case studies. Each case includes an original situation, an investigation plan and the supporting material requested in the handout, such as evidence tables, flowcharts, timelines, preservation records and a sample report outline.

Important note: The names, dates, devices and evidence entries used here are examples created to explain the forensic process. They are not results from a real examination. If this work is later supported by a laboratory exercise, the examples should be replaced with verified findings and actual screenshots.

The answers follow the assignment handout and standard digital-evidence handling practices. Background references are listed at the end.

ASSIGNMENT 2(1) / UNIQUE SCENARIO C1-EXAM-08

After-Hours Copy of an Examination Moderation Archive

Case scenario (hypothetical): At around 9:40 p.m. on 8 October 2026, the examination cell notices unusual access to a password-protected moderation archive on the departmental server, EXAM-SRV-03. The team is allowed to examine a staff workstation, WS-C14, and a 32 GB USB drive, USB-EX7, to find out whether examination files were copied.

ABOUT THIS EXAMPLE The names, dates, IDs and evidence mentioned in this case are fictional examples. The steps describe what an examiner would do and what should be checked. No real device, account, file or log has been examined for this written assignment.

What the investigation needs to establish

The first thing I would try to establish is what actually happened to the examination archive. An unusual login is concerning, but it does not by itself mean that a staff member stole a file. One possibility (H1) is that someone used WS-C14 to access the archive and copy it to USB-EX7. Another possibility (H0) is that a legitimate moderation task, backup or unrelated session explains the activity. Both possibilities need to be checked against the evidence.

For this example, I am assuming that the server kept access logs, the college has authorized a limited examination of its equipment, and the USB drive has been secured. The exact file hashes, access permissions and identity of the person using the system are not yet known. I would also check whether the computer clocks agree before using their timestamps.

Q1. Identification, collection, examination, analysis and presentation

1. Identification: I would begin with the administrator's written complaint and a clearly defined time window covering 8 and 9 October. I would list the server, workstation, USB drive and likely supporting logs, then request that important logs be preserved before they are automatically overwritten.
2. Collection: Next, I would arrange write-protected imaging of the workstation and USB drive. Server access logs and account records should be exported through approved, read-only procedures. If the workstation is still running, the team must decide whether a documented live-memory capture is justified before shutting it down.
3. Examination: I would check the acquired copies for server logins, file access records, recently used files and USB connection traces. At this stage, the aim is to identify and extract potentially relevant artifacts, not to decide who is responsible. Every extraction should point back to its original evidence source.
4. Analysis: After examining the artifacts, I would place the events in time order and compare the login account, system address, file metadata, possible copy activity and USB usage. Any differences between device clocks must be accounted for. I would also look for evidence that contradicts the copying theory.
5. Presentation: Finally, I would prepare a report explaining what was found, how the evidence was handled and how strong each conclusion is. The college committee should be able to understand the key events without having to interpret raw log entries or forensic software output.

CORE SAFEGUARD Each stage has a hand-off record. Findings about access, copying, removal and the responsible individual are stated separately because one does not automatically prove the next.

C1 / WORKFLOW AND EVIDENCE

Q2. Investigation flowchart

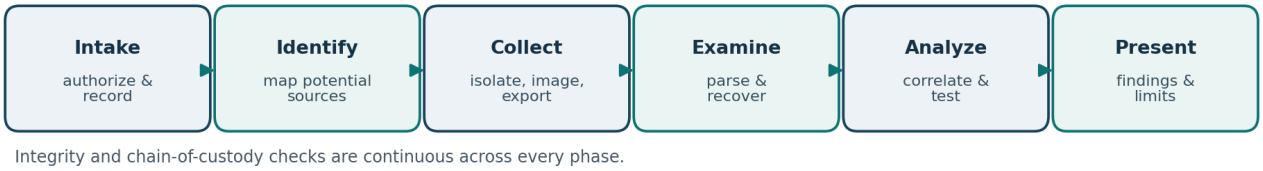


Figure 1. Proposed phased workflow for the examination archive incident.

The flowchart also has three practical stopping points. If permission to collect a source is missing, collection must stop. If an evidence seal or hash is inconsistent, that discrepancy must be investigated before relying on the copy. If a file is connected only to an account and not to a particular person, the report should say so.

Q3. Potential digital evidence and investigative value

Evidence ID / source	Potential finding	Investigative value	Principal caution
E01 / Complaint	First discovery, claimed loss, known staff access	Lead and scope	A statement is not technical proof
E02 / Server audit	Login, account, host/IP, file-open activity	Strong for server-side access	Read does not prove download
E03 / WS-C14 image	Recent files, link files, event logs	Correlates account and endpoint	Shared workstation complicates identity
E04 / USB-EX7 image	Matching archive or recovered fragments	Potential copy destination	Content can be placed by several users
E05 / Archive metadata	File name, size, timestamps, SHA-256	Object comparison / integrity	Times may be copied or modified
E06 / IAM / DHCP logs	Account privileges and IP lease mapping	Corroborating attribution	NAT, reassignment and missing logs
E07 / Door-access logs	Building presence within time window	Context only	Presence is not proof of keyboard activity

In this case, a verified copy of the exact archive on the USB would carry more weight if server logs also show access around the same time. By comparison, browser history would only offer a clue. Even a perfect SHA-256 match tells us that the file contents are the same; it does not tell us who made the copy.

Which evidence should be preserved first?

I would secure authentication, network and USB connection records early because they may be overwritten quickly. Disk images can then be examined for the relevant files and time period. Keeping the complete original acquisition is still important, even when only a few extracts are included in the final report.

Q4. Collection and preservation precautions

- Authorization and privacy: Confirm exactly what the college has permitted the investigators to examine. Student records or private staff information that has nothing to do with the suspected incident should not be reviewed unnecessarily.
- Scene documentation: Before unplugging anything, photograph the workstation, USB ports, visible messages, labels and cable connections. Note whether the machine is running, asleep or switched off.
- Preventing changes: Use a tested write blocker where suitable and examine a forensic image instead of browsing the original disk. The notes should include the software version and acquisition settings.
- Integrity checks: Calculate SHA-256 values for the collected images or exports and retain the acquisition logs. If an image contains unreadable sectors or an unexpected hash mismatch, document it rather than making the record appear error-free.
- Secure storage: Give every exhibit a unique identifier, seal it and record each handover. Stored evidence should have access controls, encryption where appropriate and a record of who opened it.
- Honest documentation: Include failed reads, clock differences and any action that may have changed the device during a live response. These details may be inconvenient, but leaving them out would weaken the investigation.

Q5. Building a defensible access timeline

To build the timeline, I would place each useful event in a single worksheet with the original timestamp, time zone, UTC equivalent, evidence ID and a short explanation of the event. Before combining sources, I would check the clock settings of the server and workstation against a trusted reference. This prevents two events from appearing to happen in the wrong order.

Hypothetical local time (IST)	Potential event	Source(s)	What it does / does not establish
08 Oct 21:36	Account MOD-02 signs in to EXAM-SRV-03	E02 + E06	Server session; not physical user identity
08 Oct 21:40	moderation_oct.zip opened/read	E02	File access; destination unknown
08 Oct 21:42	USB-EX7 device connected to WS-C14	E03	USB attach; copying still unproven
08 Oct 21:44	Matching ZIP appears on USB-EX7	E04 + E05	Strong transfer hypothesis if hashes match
09 Oct 08:25	Administrator discovers unexpected activity	E01	Discovery time, not incident time

The sample 9:44 p.m. entry is only a possible event, not something recovered from a device. In a real case, I would first confirm the archive on the USB through its hash and then check whether the workstation shows a matching transfer. The timeline becomes more convincing when different sources support the same sequence.

C1 / REPORTING

Q6. Final forensic report and non-technical presentation

Section	Content required
1. Authorization and scope	Who requested the work; legal/administrative authority; time range; exclusions.
2. Evidence register	IDs, seals, owners, sources, storage, acquisition/hash details and transfer records.
3. Methods	Isolation decisions, write blocking, imaging/export steps, tools/versions, integrity checks.
4. Verified observations	Timestamped events with exact source reference; separate facts from analyst inferences.
5. Findings and alternatives	Evidence for H1 and H0; file match, account correlation and identity limitations.
6. Conclusions	What is established / likely / not established; severity and confidence.
7. Appendices	Hash manifest, log excerpts, event schema, chain of custody, approval record, diagrams.

Example of a finding for the college committee

FINDING F-01 The hypothetical records would indicate a server login and access to the moderation archive outside approved hours. If an independently verified SHA-256 match were found on the recovered USB and USB connection times aligned, unauthorized copying would be a credible explanation. Attribution to an individual would remain conditional on separate account-ownership, access and device-use evidence.

Explaining the results to non-technical staff

I would keep the briefing simple: describe the concern, show the most important events in an easy-to-read timeline, and then explain what the evidence does and does not establish. For example, “the account accessed the archive at 9:40 p.m.” is safer than naming someone as a thief without independent proof. A hash can be described as a digital fingerprint of the file, while chain of custody is the record of everyone who handled it.

What the college should do next

The college should restrict access to the affected examination material and, after preserving evidence, review account permissions and potentially exposed passwords. Interviewing relevant staff may help clarify the timeline. Any security repair should be planned so that important logs or files are not accidentally deleted.

C1 / LEGAL AND ANALYTICAL CLOSE

Q7. Legal and ethical considerations in a college environment

This investigation takes place in a college, where the evidence may contain examination papers and personal information. The investigators therefore need written authority, a limited purpose and careful handling of private material. Having permission to monitor a college server does not automatically give permission to read unrelated private messages. Applicable Indian cybercrime, privacy and electronic-evidence requirements should be checked with the appropriate authority.

- Fair treatment: Affected staff or students should have an opportunity to explain the circumstances. The team should also preserve evidence that points away from the original suspicion.
- Confidentiality: Keep examination questions, personal identifiers and unrelated files accessible only to authorized people. Redact unnecessary personal details from committee copies.
- Proportionality: Do not search more widely than necessary, alter original files or circulate raw evidence through personal email or messaging accounts.
- Retention: Record who accessed the evidence, how long it must be retained and how verified copies may be provided to an authorized authority.

My assessment of the case

I would separate two questions in the final assessment: was the archive accessed or copied without permission, and can the responsible person be identified? Server logs might establish the access, while a matching file and USB traces could support the copying allegation. However, the account owner is not automatically the operator. Shared credentials, remote access or a compromised session could produce similar evidence.

Claim	Minimum support expected	Residual uncertainty
File accessed	Authenticated server audit showing access to named archive	Account misuse or service action
File copied to USB	Matching complete contents and contemporaneous endpoint/USB artifacts	Who placed it there and when
Named user operated device	Independent endpoint login, physical/remote presence and credible witness corroboration	Shared access, spoofing or compromise

At this stage, the case is still an allegation. The proposed process would help the college reach a careful decision without compromising examination confidentiality or unfairly blaming a person before the evidence has been checked.

ASSIGNMENT 2(2) / UNIQUE SCENARIO C2-HD07-11

Shared Helpdesk PC and a Disguised Maintenance Executable

Case scenario (hypothetical): A helpdesk computer, HD-07, is shared by several members of the hostel administration. The computer appears to contain an executable named update_service.exe in a hidden folder, along with encrypted duty-roster documents and browser activity. Investigators want to know which account created or used these files, while remembering that staff and automated services may share access to the machine.

ABOUT THIS EXAMPLE The names, dates, IDs and evidence mentioned in this case are fictional examples. The steps describe what an examiner would do and what should be checked. No real device, account, file or log has been examined for this written assignment.

Q8. Hardware and software inventory before analysis

Item	What the examiner records	Why it matters
Physical device	Make/model, asset tag, serial, BIOS/UEFI, photos, peripherals	Provenance and boot context
Storage	NVMe/SATA model, capacity, serial, partitions, sector size, encryption	Correct image source and limitations
Memory / boot	RAM size, power state, boot method, external media	Possible volatile artifacts
OS	Windows edition/build, install path, accounts, locale, clock/time zone	Parsing and timestamp context
Filesystem	NTFS volumes, mount points, ACLs, USN journal, $MFT availability	Recoverable activity metadata
Security apps	AV/EDR, encryption tool, browser builds, scheduled tasks	Benign or suspicious execution clues
Network	Host/IP/MAC, domain membership, VPN/proxy settings	Cross-system attribution

Before looking into the suspicious file, I would document the computer itself: the drive model and serial number, available memory, connected media, operating system, installed programs, user accounts and file system. Most of this can be checked from scene notes and a read-only image. I would not boot the original machine just to fill in a missing inventory field.

Q9. What storage, OS, filesystems and metadata contribute

The storage drive may hold ordinary files, deleted file remnants and metadata. On a Windows system, NTFS records such as the Master File Table ($MFT) can help locate a file, while account profiles and application records may show how it was accessed. Creation and modification times, file paths, hashes and ownership are useful clues, but they must be interpreted together. One timestamp cannot prove that someone executed the program.

For update_service.exe, I would compare the folder location and filesystem record with browser download history, recent-file shortcuts, execution traces and scheduled tasks. I would also check whether the file was renamed or moved. If the system uses an SSD, some deleted data may already be unavailable because of TRIM.

C2 / HIDDEN AND ENCRYPTED CONTENT

Q10. Hidden files, encrypted files and safe evidence handling

A hidden file is not necessarily malicious; the hidden attribute normally just prevents a file from appearing in everyday folder views. I would examine hidden and system directories, alternate data streams and deleted entries from the image, keeping the original attributes unchanged. A confusing extension or filename would be a reason to inspect a file, not proof of deliberate concealment.

Encryption creates another difficulty because the investigator may see that a file exists without being able to read its contents. Whether the protection is on a document, archive or entire disk, any available recovery key must be used only with proper permission and on a copy. Even without decryption, filenames, sizes, application traces and metadata may help answer the case questions. I would not claim to know what an encrypted document contains until it has been lawfully opened.

- Record each relevant file's full path, drive or volume identifier and the image from which it was extracted.
- Give extracted files their own SHA-256 hashes, and record any steps that change their representation, such as decryption or decompression.
- If a recovery password or key is legally provided, use it on a duplicate image and preserve both encrypted and decrypted versions with separate identifiers.
- Store passwords and keys securely, keep a record of who accessed them and avoid exposing unrelated duty-roster information.

Illustrative examination tests

Observation / test	Interpretation if confirmed	Competing explanation
update_service.exe has hidden attribute and runs from user Temp	Potential evasion / staging	Installer temporarily expands files
File hash matches a known malicious or approved binary	Meaningful classification clue	Hash does not identify operator
Encrypted roster created within account session	Possible account interaction	Background sync or automation
Browser download URL matches file hash	Acquisition route supported	Another user downloaded on shared profile

Before putting a hidden or encrypted file into the report, I would ask whether it actually helps answer the allegation. Personal material that is unrelated to the case should remain preserved in the evidence image but should not be disclosed in the report.

C2 / ATTRIBUTION MATRIX

Q11. Evidence-linkage matrix: file to user account

The table below illustrates how I would test a possible link between a file and a Windows account. The listed people, file paths and times are examples, not findings from HD-07.

Source artifact	Illustrative linkage	What it may support	Strength / caveat
NTFS $MFT / USN	C:\Users\staff.b\AppData\Local\Temp\update_service.exe, creation ~14:06	Object existed and changed in profile area	Medium: path ownership not human proof
Browser downloads	User profile staff.b: URL to update_service.exe at ~14:04	Browser acquisition occurred	Medium: browser profile may be shared
Windows logon	Session for staff.b active ~13:50–14:35	Account session overlaps write window	High for account, not operator
Execution trace	Prefetch/other execution indication ~14:08	Binary may have executed	Varies by artifact and OS configuration
Task Scheduler	Task “HostelSync” references same executable	Persistence or approved scheduled task	Context-dependent, review creator
Recent file artifacts	Link to encrypted roster from same account	Interaction with target document	Medium; shortcuts can be indirect
AV/EDR / events	Process/file-detection event tied to SID and process	Additional process context	Strong if audit reliable and complete
HR/room logs	Staff.b signed duty roster that shift	Possible user availability	Supporting only; another user possible

I would rely on stable identifiers when matching records: file hashes are more dependable than filenames, account SIDs are more reliable than display names, and timestamps must be converted to the same time zone. Even after these checks, an artifact can connect activity to an account without identifying the person who used it.

How the traces could fit together

One possible sequence is a browser download at 2:04 p.m., file creation at 2:06 p.m., execution at 2:08 p.m. and access to an encrypted roster at 2:12 p.m. I would only connect those events if the corresponding artifacts exist and agree. A background software updater might otherwise explain part of the same sequence.

C2 / ILOOKIX WORKFLOW

Q12. ILookIX acquisition/import and analysis workflow

ILookIX is an older forensic examination program. Its menus and supported image formats can vary with the release, so I would describe the checks an examiner must make rather than inventing screenshots. If ILookIX is unavailable, a tested alternative could be used, but the final report should name the substitute and explain that decision.

1. Receive the authorized evidence, check the exhibit identifiers and verify the hash of the disk image. Importing an image into software is not the same as acquiring the original disk.
2. Set up a new examination case and record the examiner's name, the ILookIX version, the computer clock and the relevant settings.
3. Import the verified image, note the partitions and file systems that the tool recognizes, and record any error messages. If a different image format is needed, keep both the original and converted copies.
4. Look for hidden or deleted files, encryption indicators and files whose contents do not match their extensions. Examine update_service.exe and compare its hash with a trusted copy if one exists.
5. Review user profiles, SIDs, browser downloads, shortcuts, execution traces and scheduled tasks. Record the words, hashes and dates used for each search so it can be repeated.
6. Export relevant artifacts with their original paths, timestamps, source-image details and hashes. Treat every export as a separate, traceable evidence item.
7. Compare the results with independent logs and account information, then prepare an evidence index that another examiner can check.

EXPECTED OUTPUTS A verified image import record; partition/user-account inventory; filtered file listing; executable hash and metadata; account-artifact links; export manifest; and a transparent analysis log. These are desired outputs, not observed outputs of a tool run.

Validation checkpoint

Before drawing conclusions, I would reopen a few exported files in another viewer and verify their hashes. I would also list anything the software could not parse, such as encrypted content or unsupported event-log records, rather than treating missing results as proof that nothing happened.

C2 / CONCLUSION AND LIMITATIONS

Q13. Direct versus supporting links

Direct evidence is an artifact that records the activity being investigated. For instance, a reliable process log with a Windows account SID may directly connect program execution to that account. It still does not prove which human was at the keyboard. A file found in someone's profile, a room schedule or a matching IP address is usually supporting evidence because each has several possible explanations. A genuine admission could be stronger evidence of a person's involvement, but its authenticity would also have to be checked.

Example claim	Direct account-linked artifact	Supporting context
A particular account ran the executable	Trusted process execution audit keyed to account SID and executable identity	Browser downloads, same-profile file path
Executable accessed the roster	Credible process/file-open audit tying process to object	Timestamps and a recent-documents shortcut
A named person operated the machine	Independent authenticated identity mechanism plus corroboration	Shift roster or badge presence alone

Q14. Limitations and alternative explanations

- Shared access: A colleague could have used an unlocked session or a remote-control tool, even though the account belonged to someone else.
- Unreliable time order: Copied files, restored backups, delayed writes and incorrect clock settings may change how the events appear in the timeline.
- Background software: Updates, antivirus scans, scheduled tasks and backups can create files or trigger programs without a user clicking them.
- Missing evidence: Deleted browser records, overwritten logs, disabled auditing and SSD TRIM may hide useful context.
- Mistaken suspicion: A hidden folder or strange executable name is worth checking, but it may still belong to a legitimate maintenance program.

Balanced conclusion for scenario C2-HD07-11

If browser, NTFS and execution evidence all point to one Windows account, I could reasonably report an account-level connection. I would not name the account owner as the operator until shared access, automation and remote sessions have been considered. The report should identify the exact executable using its hash and state clearly whether its behavior was actually tested. Since this is a planned examination, there is no confirmed suspect at present.

ASSIGNMENT 2(3) / UNIQUE SCENARIO C3-ROB-L11

Seizure of a Robotics Sponsorship Laptop and USB Drive

Case scenario (hypothetical): A robotics-team laptop, ROB-L11, and a 64 GB USB drive are collected after a confidential sponsorship proposal is reportedly shared outside the team. The laptop is found switched off inside a locked innovation-lab cabinet. The priority is to collect both devices safely and make verified images before anyone begins searching the files.

ABOUT THIS EXAMPLE The names, dates, IDs and evidence mentioned in this case are fictional examples. The steps describe what an examiner would do and what should be checked. No real device, account, file or log has been examined for this written assignment.

Q15. Collection and transport: safe step-by-step protocol

1. First, confirm the written permission to collect the devices and identify the people responsible for the seizure. Note the case number and the exact location covered by the approval.
2. Photograph the cabinet, laptop, attached USB, charger, ports and any visible labels or damage. Record who could access the cabinet.
3. Check that the laptop is actually powered off. Do not start it or connect it to a network. If it is on unexpectedly, follow an approved live-response decision process instead of guessing.
4. Note the connected cables and devices before disconnecting them. Package the USB as its own exhibit and record the port from which it was removed.
5. Record make, model and serial numbers for both items. Attach tamper-evident labels and give each device a separate exhibit ID.
6. Place the devices in suitable protective packaging and keep them away from moisture, shocks and unauthorized handling during transport.
7. Record each handover in the chain-of-custody form and store the sealed exhibits in a secure intake area when they reach the laboratory.

LAPTOP-SPECIFIC CAUTION Do not assume power state from a dark screen. Sleep/hibernation, hardware encryption, battery removal and modern SSD behavior may change the correct acquisition decision. Record the chosen method and why the risk was acceptable.

Custody identifiers

I would use ROB-01 for the laptop and ROB-02 for the USB. Their forensic images would be ROB-03 and ROB-04. If an examiner creates a working copy, it should have its own linked identifier, such as ROB-03-W1. This makes it possible to see which item every later result came from.

C3 / ACQUISITION OVERVIEW

Q16. Purpose of physical acquisition and imaging copies

Physical acquisition means copying the addressable sectors of a storage device, not just the files visible in File Explorer. It can include active files, filesystem structures and remnants in unallocated space. I would normally examine the verified image instead of the original because a normal program might write temporary files or change metadata. Keeping the original untouched also makes it easier for another examiner to repeat the work.

However, a physical image does not guarantee that every old byte will be available. SSD TRIM, encryption, unreadable sectors and some firmware-controlled areas may prevent full recovery. I would state precisely whether the acquisition covered a whole disk, one partition or only selected files.

Q17. IXImager-based imaging workflow

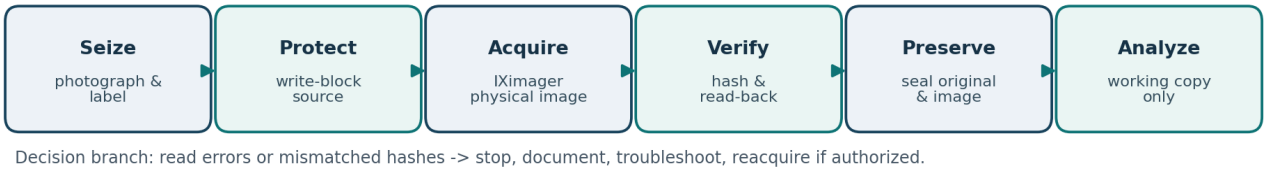


Figure 2. Physical-acquisition stages and exceptional-condition branch.

IXImager is an older Linux-based acquisition tool associated with ILook. For this assignment, I have outlined a workflow based on its documented purpose, rather than pretending that the tool was run. In a real laboratory, I would first check which version is available and whether it supports the evidence devices and intended image format.

Gate	Procedure and recorded evidence
Source identification	Record drive model/serial, interface, capacity, partition signs and selected source address; compare against scene inventory.
Write protection	Connect through verified read-only hardware or documented equivalent; test blocker status; prohibit source mounting as read-write.
Destination	Prepare clean, adequately sized evidence storage with separate case directory and access permissions.
Imaging	Choose physical source and supported image/container option; capture tool/build, operator, start/end time, read errors and output split files.
Validation	Record tool verification, independent image read-back, size/sector coverage and cryptographic hashes; retain logs.
Preservation	Seal originals, encrypt/archive master image, create a derivative working copy and register all transfers.

Operational acquisition checklist

I would acquire the laptop and USB separately. They may be connected at the scene, but each one has its own device serial number, size, imaging log, hash and custody history. Imaging the laptop cannot be treated as an image of the detached USB drive.

1. Check that the write blocker or read-only interface really prevents writes, and save the test or equipment record.
2. Confirm the source drive and its capacity before starting. This is essential because selecting the destination by mistake could destroy evidence.
3. Start imaging and keep the tool log. A progress bar reaching 100% is not enough; check the final status and any bad-sector warnings.
4. Use the supported verification procedure to compare the image with the intended source range. Record exactly what was verified.
5. Store the master image securely and calculate hashes for the relevant image or container files and segments.
6. Create a separate working copy. Check it before examination and document any later exports or format conversions.

Q18. How hashes demonstrate evidence integrity

A hash is useful because the same bytes should produce the same SHA-256 value. I would calculate and record hashes at defined points: acquisition, verification, copying and later access. The comparison must be made between the same type of data. For example, the hash of a compressed evidence container will not normally equal the hash of the raw disk data inside it. Both may be correct if their meanings are documented.

Integrity checkpoint	Field to record	Decision rule
Before / at acquisition	Source read hash, if available, with sector scope and errors	Record baseline and limitations
After image creation	Hash of reconstructed raw data or tool verification result	Compare like with like
Master archive	SHA-256 of each image container and segment	Manifest immutable master files
Before/after analysis	SHA-256 of read-only image/working copy	Investigate any unexplained change

EXCEPTION HANDLING If sector reads fail, no honest examiner should invent a full source/image equality result. Document unreadable logical block addresses, retries and filler policy; state that verification pertains to the successfully captured and defined content.

Q19. Role of the ASB forensic evidence container

The ASB format is associated with older IXImager forensic images. It was designed to keep acquired data with tool-related authentication information in a container that compatible software could open. I would use it only after checking support for the specific tool and version. The file extension alone cannot guarantee integrity or long-term compatibility.

Along with the ASB image, the team needs a case and device identifier, source serial number, examiner name, acquisition time and time zone, imaging-tool version, segment list, hashes, verification results, error logs, custody entries and storage details. If another image format is created for compatibility, both representations and the conversion should be documented.

Q20. Data carving when filesystem information is absent

Data carving is useful when normal file-system records are missing or damaged. Instead of relying on a file name or folder entry, it searches the raw image for recognizable file structures, such as a header or trailer. It may recover parts of a deleted presentation, picture or archive, but it usually cannot establish the original folder or owner by itself.

I would run carving only on a verified working image and save the offsets, signatures and settings used. A recovered file should be checked to see whether it opens correctly and whether it is complete. Fragmentation, overwriting, SSD TRIM and encryption may leave only partial content or nothing recoverable.

Q21. Date/time uncertainties and documentation

Problem	Examiner action
Clock offset or drift	Record device BIOS/OS time and compare with trusted time; quantify offset and uncertainty.
IST versus UTC	Retain raw timestamp plus source zone; convert consistently to UTC (+05:30 for IST).
Filesystem semantics	Differentiate NTFS creation/modify/access/MFT-change fields and file-copy effects.
Precision and log delays	Describe rounding, caching, scheduled tasks, delayed writes and gaps.
Time manipulation	Inspect time-service events and other independent records; do not retroactively “fix” source records.

C3 / CHAIN OF CUSTODY

Q22. Chain-of-custody record: ROB-01 laptop

This is a sample custody form showing how the laptop should move from the person who collects it to the laboratory examiner. The example names and times must be replaced with real signatures and observations if a practical seizure takes place.

Date/time (IST)	From -> to / role	Location and purpose	Seal / condition / sign
10 Oct 2026, 10:15	Campus security -> Collector A	Innovation lab; initial seizure	Seal SL-101 applied; both sign
10 Oct 2026, 10:45	Collector A -> Transport B	Secure movement to lab	SL-101 visually intact; both sign
10 Oct 2026, 11:10	Transport B -> Evidence officer C	Forensic locker intake	SL-101 intact; vault entry recorded
11 Oct 2026, 09:05	Evidence officer C -> Examiner D	Authorized imaging session	Seal opened/resealing logged
11 Oct 2026, 15:30	Examiner D -> Evidence officer C	Locker return after imaging	New seal SL-102; both sign

Mandatory fields for each actual record

A real custody entry should show the exhibit ID, make and serial number, person handing it over, person receiving it, reason for transfer, date and time with time zone, location, seal condition and signatures. If the seal is opened and replaced, both actions must appear in the record. The sample entries here are only illustrations.

Acquisition integrity statement for the final report

The final report could explain the procedure in these words: “The laptop and USB will be documented and acquired separately through read-only procedures. The image and acquisition logs will record the exact source range, errors and verification results. The originals will be secured, and examination will take place on working copies.” This wording is intentional because no actual acquisition was performed for this written case study.

If the disk is encrypted and there is no authorized way to obtain its key, I would say that some evidence could not be examined. A transparent limitation is more defensible than an undocumented step that risks changing the original device.

ASSIGNMENT 2(4) / UNIQUE SCENARIO C4-GRANT-R04

Deleted Research Grant Ledger Recovered from Shadow Copies

Case scenario (hypothetical): A finance-office computer, FIN-R04, appears to have lost a research-grant spreadsheet after an unusual late-night edit. The investigators already have a verified Windows image containing NTFS records and Volume Shadow Copies. Their task is to find out whether an older copy can be recovered and whether the traces really show a deletion.

ABOUT THIS EXAMPLE The names, dates, IDs and evidence mentioned in this case are fictional examples. The steps describe what an examiner would do and what should be checked. No real device, account, file or log has been examined for this written assignment.

Q23. Evidence selection and triage before full analysis

I would begin with a few focused questions: Which version of the grant ledger existed before the incident? What changed, when did it change, and was the file deleted or simply moved? I would preserve the full disk image but search only the relevant folders, filenames, dates and activity records to avoid examining unrelated finance information.

1. Check the source image ID and hash, the volumes present, the system clock and whether shadow copies are available.
2. Identify the likely ledger names and locations, Office recent-file entries, NTFS $MFT and journal records, recycle-bin entries, and relevant browser or application activity.
3. Group potential evidence into original content, system activity, recovered copies and supporting records. “Suspicious” should only mean “needs closer checking.”
4. Use hashes to recognize duplicate files and exclude ordinary system data from the review, while keeping a record of what was filtered out.
5. Record the exact dates, keywords and filters used, especially because several shadow copies may contain versions of the same document.

Q24. Evidence classification scheme

Class	Illustrative member	Treatment / investigative role
User-created	GrantLedger_Q3.xlsx, locally drafted letters	Primary business content; inspect approved versions
System/app	NTFS metadata, Office recovery, Windows event records	Time and software behavior; parser-validation required
Potentially suspicious	Unusual renamed ledger or cleanup script	Review behavior and alternative intent before labeling
Deleted/recovered	Recycle Bin entries, VSS copy, unallocated fragment	Recoverable historic content with origin recorded
Supporting/context	Cloud-sync logs, browser history, user sessions	Corroborate transfer or account timing

Q25. Using event analysis to reconstruct activity

Event analysis helps turn scattered records into a clear sequence. In this case, NTFS may show file changes, Office records may show a recent edit, Windows logs may indicate which account was signed in, and cloud-sync traces may show another copy. I would combine these details in a timeline while keeping the raw times and their UTC conversions.

A useful timeline also explains what each artifact can actually prove. A “last modified” date does not describe every edit, and a browser visit does not confirm that an upload succeeded. A deleted entry may result from a normal application save process. These differences should be visible in the final interpretation.

Q26. Cloud Analysis and Lead Analysis

Method	Question it answers	Representative records	Caution
Cloud Analysis	Was the ledger synchronized, shared, versioned or uploaded?	Authorized provider audit, revision history, object IDs, access logs	Cloud timestamps and retained versions differ by service
Lead Analysis	Which clue deserves the next validated search?	Filename variants, username, USB ID, recipient domain	A lead is a hypothesis until corroborated
Combination	Could a local deletion coexist with a cloud-retained copy?	VSS ledger version + sync/upload time + cloud revision	Same name or size does not prove identical content

Example lead workflow

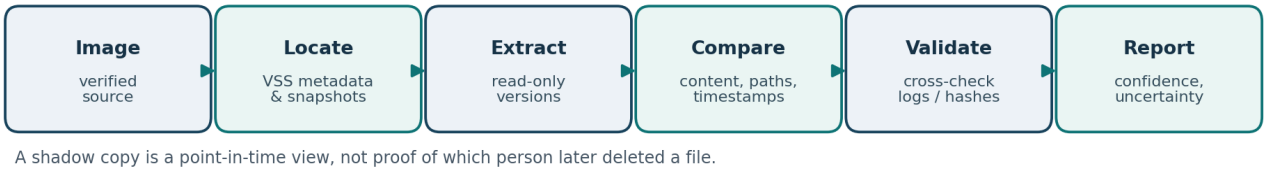
Suppose an older ledger contains the project code RG-41. I would search for that code in the approved image, check related browser or transfer traces, and then ask for a narrowly scoped cloud audit export if needed. I would record a negative search result too. That is more reliable and respectful of privacy than opening every financial document.

Confidence vocabulary

To make conclusions easy to follow, I would use “observed” for an artifact that has been verified, “corroborated” when two independent sources agree, “plausible” when an explanation has support but alternatives remain, and “unresolved” when there is not enough evidence. Numerical confidence scores would not be appropriate without a defined method.

C4 / VOLUME SHADOW COPIES

Q27. Volume Shadow Copy discovery and recovery workflow



A shadow copy is a point-in-time view, not proof of which person later deleted a file.

Figure 3. Read-only shadow-copy recovery and validation process.

Volume Shadow Copy Service (VSS) can retain an earlier state of a Windows volume. If a spreadsheet changed or disappeared after a snapshot was created, the snapshot might still contain its earlier version. A snapshot is not a recording of everything that happened between two times, however, and older snapshots may already have been deleted.

1. Open the verified disk image in read-only mode and identify the NTFS partitions and any stored shadow-copy metadata.
2. List the available snapshots and record their identifiers, source volumes, capture times and any errors reported by the forensic tool.
3. Search for the grant ledger and related browser records in both the current filesystem and each relevant snapshot.
4. Export every useful version separately, with its original path, snapshot identifier, metadata and SHA-256 hash.
5. Compare versions using hashes and a careful content review. A different hash means the byte content differs, but the report should explain the actual changes.
6. Cross-check the versions against $MFT, USN, Office, recycle-bin or approved cloud records to test whether editing, moving or deletion took place.
7. Save the extraction log, check exported documents in a second viewer and state any limitations caused by missing or damaged snapshots.

CRUCIAL LIMITATION A file present in a 17:00 snapshot but absent from the current volume establishes that the point-in-time states differ. It does not, without more evidence, show the exact deletion time, deleting process, or the person responsible.

C4 / WORKED TIMELINE AND VALIDATION

Q28. Hypothetical file life-cycle timeline

The timeline below is a made-up example, written in IST to make the sequence easier to read. In a real case, I would also save the UTC equivalent and note any clock uncertainty.

Time / artifact	Illustrative state	How it would be checked
06 Oct 09:20 / NTFS	GrantLedger_Q3.xlsx created	File record, path and project code
07 Oct 15:10 / Office + NTFS	Ledger modified: grant totals revised	Two contents or Office revision data
07 Oct 17:00 / VSS-A	Snapshot includes revised ledger	Snapshot ID + export hash
07 Oct 18:05 / sync log	Candidate copy to a team folder	Provider action/result plus object ID
08 Oct 21:18 / $UsnJrnl	Rename/delete-type filesystem event	USN reason, matching file reference
09 Oct 11:40 / forensics	Older ledger exported from VSS-A	Read-only export + chain/hash

I would be careful not to read more into an NTFS journal entry than it shows. An Office application may create a temporary file and rename it during an ordinary save. A file may also have been moved instead of deleted. The timeline should be checked against those possibilities.

Q29. Recovered-artifact validation checklist

- Source image verified against its preserved manifest before and after extraction.
- VSS snapshot identifier, volume, capture time and extraction tool/build preserved.
- Exact recovered path, file record/offset when available, and object metadata noted.
- Exported recovered item assigned a unique exhibit ID and SHA-256 digest.
- File signature and format checked independently of extension; internal structures open without silent repair.
- Content compared against current copy and any trusted business copy; similarity stated precisely.
- Timestamps normalized but raw values retained; creation within snapshot not confused with snapshot time.
- Any parser warnings, corruption, partial fragments or missing proof disclosed.

C4 / UNCERTAINTIES AND CONCLUSION

Q30. Errors and competing explanations for deleted evidence

- Snapshot timing: A snapshot only shows the state at capture time. The spreadsheet could have been edited much earlier.
- Normal Office behaviour: Temporary files, rename-on-save operations and cloud synchronization may resemble intentional deletion.
- Timestamp differences: Incorrect clocks, time-zone conversion, imported files or retained metadata may put events in the wrong order.
- Who used the account?: A logged-in account could have been shared, accessed remotely or used by a scheduled service.
- Missing copies: Old snapshots can be deleted by cleanup or storage limits; TRIM and overwriting can also remove recoverable data.
- Uncertain recovery: A carved document might have lost its original filename or folder and could be mistaken for a similarly named document.

Case synthesis and validation decision

If the shadow-copy records and the exported file are validated, I could report that a particular version of the ledger existed when that snapshot was taken. A later missing file and supporting NTFS records might suggest deletion, but neither fact alone shows a dishonest intention. Identifying the person responsible would require separate login, process or other supporting records.

Finding category	Reporting language
Confirmed if validated	“An earlier version of GrantLedger_Q3.xlsx was recovered from snapshot VSS-A.”
Corroborated if supported	“A filesystem change event and a second source align with later removal or replacement.”
Not established	“The current materials do not identify who deliberately removed the file.”

Closing evidence package

The final evidence package should contain the classification table, snapshot register, original and recovered file details, version comparison, event timeline, validation checklist and hashes. I would include only the finance information needed to explain the findings and protect any unrelated confidential details.

ASSIGNMENT 2(5) / UNIQUE SCENARIO C5-PORTAL-09

Compromised Campus Poster-Portal Account Across Five Sources

Case scenario (hypothetical): A third-year student notices that an unfamiliar abstract was submitted through a campus sustainability poster portal using their account around 11:12 p.m. on 9 October 2026. With suitable permission, investigators can examine the student's Windows laptop, browser data, relevant messages, smartphone and the portal's cloud audit records. The main question is whether the student made the submission or someone else used the account.

ABOUT THIS EXAMPLE The names, dates, IDs and evidence mentioned in this case are fictional examples. The steps describe what an examiner would do and what should be checked. No real device, account, file or log has been examined for this written assignment.

Q31. Windows and other operating-system evidence

I would first look at the Windows laptop for login events, account profiles, browser data, downloads, recent files, system time settings and relevant NTFS records. If the activity involved a Linux or macOS device, similar evidence might be found in login logs, application histories, filesystem metadata and browser profiles. Each operating system needs its own validated collection approach.

It is important not to treat all logins as the same thing. Being logged into Windows, opening a browser profile and having a valid portal session are three different observations. The final report must say exactly which layer the evidence identifies before drawing a conclusion about the student.

Q32. Browser artifact value and careful interpretation

Browser history might show that the portal was visited, while cache, downloads, saved logins, form information and extensions may provide further context. A submitted-page address does not necessarily mean the server accepted a file. Browsers may preload pages, synchronize history across devices or reuse a stolen session cookie, so I would check any browser clue against the portal's own logs.

Browser artifact	Potential value	Main caveat
History + visit timestamp	Page access opportunity	Automated/preloaded visit possible
Cache / form remnants	Candidate content of submitted abstract	May be stale or incomplete
Download list	File obtained before alleged submission	Downloaded file not necessarily uploaded
Session token / cookie	Authentication context for portal	Token possession not proof of person

C5 / COMMUNICATIONS AND MOBILE

Q33. Email and messaging evidence and correlation

Email and chat may contain login warnings, password-reset notifications, verification codes, attachments and conversations about the abstract. Message headers, message IDs and provider delivery records are generally more useful for authentication than a screenshot showing only a display name. I would collect only the relevant conversations and preserve their original metadata.

For example, a password-reset email could be matched with a provider record of a reset; an attached abstract could be compared by hash with the uploaded document; and a verification-code message could be compared with a login attempt. Sending, receiving and reading a message are different events, so I would not assume one proves the others.

Q34. Mobile-phone forensic considerations and challenges

- Initial condition: Photograph the phone screen, battery and connection status. Any decision to disconnect the network should follow a documented, device-specific plan.
- Permission and privacy: Establish whether the phone belongs to the student or the college and collect only the data covered by consent or another lawful authority.
- Technical limits: Device encryption, app protections and the absence of a passcode may restrict the team to logical data or prevent collection altogether.
- Useful material: Where accessible, check account identifiers, portal-app or browser activity, authentication prompts, notifications and relevant downloaded files.
- Interpretation: A notification may be delayed or synchronized from elsewhere. Even an approved login prompt does not necessarily establish that its owner knowingly authorized the action.
- Handling: Record the extraction method, software version, collected scope, device identifiers and integrity checks. Do not browse through unrelated personal applications.

Mobile verification question

If the portal shows that an MFA request succeeded at 11:10 p.m., I would ask which device received it and what exactly was approved. I would also check whether the laptop already held a valid session. MFA success can support an account-level event, but it does not by itself identify the person involved.

C5 / INTERNET AND CLOUD

Q35. Relevant Internet and cloud evidence

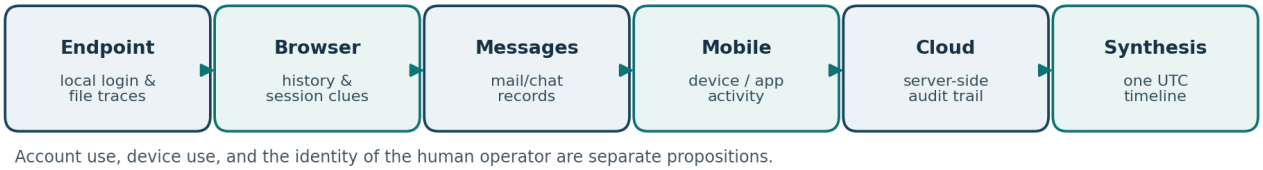


Figure 4. Multi-source correlation path; a shared timeline does not collapse source uncertainty.

The portal provider may have login results, source IP addresses, user-agent strings, upload times, file hashes, object versions and session-revocation records. Supporting network evidence might include campus DHCP, VPN or proxy logs. I would request those records using the proper authority and check the provider's retention limits.

Server records are particularly helpful because they can confirm what the portal accepted, instead of only what a user's device displayed. Still, an IP address is not a person. Shared Wi-Fi, VPNs, NAT and mobile carrier networks can make address-based attribution unreliable. Every provider export should include its scope, export time, time zone and responsible administrator.

Priority and minimization

Preserve first	Scope	Reason
Portal auth + object audit	08–10 Oct, account and object IDs	Rotating records may be lost
Password reset / MFA events	Target account, same window	Tests account compromise
Relevant endpoint/browser state	Authorized profiles and dates	May clarify device session
Cloud object versions	Submitted abstract and revisions	Authoritative result of upload
Network allocation records	IP/time mapping, precise zones	Contextual source correlation

The goal is to obtain original, traceable exports rather than copying information informally from a dashboard. I would hash exported files, preserve how they were produced and include only the records needed for the incident.

C5 / CROSS-SOURCE CORRELATION

Q36. Cross-source artifact matrix: five independent evidence classes

This matrix is a worked example showing how evidence from five sources might support or challenge each other. The events have not been collected from real accounts or devices. All displayed times are in IST; actual analysis should also record UTC and any clock differences.

Source & illustrative time	Hypothetical artifact	Correlated item / meaning	Limitation
Windows endpoint / 22:58	Student account unlock on LAP-22	Computer session before portal activity	May not be incident actor
Browser / 23:05	Poster portal login page and abstract form cache	Session/possible form preparation	Cache may predate submission
Email / 23:07	Provider notification: “new sign-in”	Candidate authentication activity	Verify headers and provider audit
Mobile / 23:10	Authentication-app approval entry	MFA request linked by transaction ID	Approval context still ambiguous
Cloud portal / 23:12	Object POST / submission ID PB-417	Server accepts an upload; file hash needed	IP/account do not identify a human
Network / 23:12	DHCP lease for public-IP gateway	Potential route from campus network	NAT may represent many users
Chat / 23:18	Message containing purported abstract title	May connect content to communication	Account could be impersonated

Correlation keys and alternative hypotheses

The strongest comparisons use stable values such as a portal transaction ID, uploaded-file hash, device identifier or verified account SID. Similar filenames, IP addresses and close timestamps are only supporting clues. I would test three explanations: the student submitted the abstract, another person used stolen credentials or an existing session, or a shared device or authorized workflow caused the activity.

C5 / CHALLENGES AND FINAL JUDGMENT

Q37. Challenges for distributed digital forensics

1. Different permissions: Cloud providers and personal devices may require separate legal or administrative authorization.
2. Short-lived evidence: Provider logs, browser sessions, app notifications and chats may be deleted or rotated.
3. Encryption: Modern phone and computer security features may limit the evidence a forensic tool can obtain.
4. Uncertain identity: An account name, IP address or MFA approval does not always identify the human using the service.
5. Different clocks: Providers and devices record time in different zones, sometimes with delays or incorrect clock settings.
6. Changing data formats: Browser, messaging and cloud export formats can change, and not every tool can parse them.
7. Too much unrelated data: An investigator must avoid reviewing private messages and files that do not concern the incident.
8. Chain of custody: Screenshots or exports without a proper collection record can be difficult to verify later.
9. Duplicate traces: One cloud event may synchronize to many devices, making a single activity appear to have several sources.

Q38. Final investigative conclusion with evidentiary tiers

Tier	Potential conclusion in scenario C5	What remains necessary
Direct platform evidence	A validated server audit would show the account accepted submission PB-417 at 23:12.	Original provider log, object record and query provenance.
Corroborating evidence	Browser cache, security email and mobile approval may align with the portal event.	Device/account ownership, authentic headers and time normalization.
Unresolved questions	Which person controlled the session, how login was obtained and whether approval was voluntary.	Independent interviews, safe credential review and any lawful additional records.

My provisional conclusion would be limited to what the evidence can support. If the portal's own logs confirm the upload, I could say that the student's account was used for the submission. I could not say the student personally uploaded the abstract without further corroboration. The report should explain all reasonable alternatives and recommend proportionate steps, such as preserving logs, then reviewing the session and MFA settings.

When the laptop, browser, messages, phone and portal agree on the sequence, confidence in the event timeline increases. That agreement still needs to be separated from the harder question of who performed the action and why.

REFERENCES / APPLICABLE BACKGROUND GUIDANCE

Reference Notes and Submission Checklist

The assignment handout is the main source for the questions and deliverables. The following references provide background on digital-evidence preservation, the older IXImager terminology and Windows Volume Shadow Copies. They support the proposed workflows but are not a substitute for approved lab procedures.

- [1] NIST SP 800-86, Guide to Integrating Forensic Techniques into Incident Response. https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-86.pdf
- [2] NISTIR 8387, Digital Evidence Preservation: Considerations for Evidence Handlers (2022). https://doi.org/10.6028/NIST.IR.8387
- [3] National Institute of Justice, Test Results for Digital Data Acquisition Tool: IXimager Version 2.0 (2007). https://nij.ojp.gov/library/publications/test-results-digital-data-acquisition-tool-iximager-version-20-feb-01-2006
- [4] Microsoft Learn, Volume Shadow Copy Service (VSS). https://learn.microsoft.com/en-us/windows-server/storage/file-server/volume-shadow-copy-service
- [5] Forensics Wiki, IXimager and IXimager file formats (historical summaries). https://forensics.wiki/iximager/ and https://forensics.wiki/iximager_file_formats/

Final review before submission

- Answers to all five case studies and all 38 questions have been included in the same order as the handout.
- Each assignment uses a distinct case scenario, with assumptions clearly marked.
- The required diagrams, evidence tables, timelines, report outline, and sample chain-of-custody form are included.
- Sample evidence, times and custody entries are clearly identified as examples rather than actual tool output.
- The answers explain forensic integrity, attribution limits, alternative hypotheses and responsible evidence handling.
- The cover page contains the required student details and submission date; review the final version before handing it in.
- If a separate practical demonstration is requested, add results collected with permission and label them accurately.

As stated in the handout, the submission deadline is 16 October 2026, before 4:00 p.m. Students already approved for the seminar are exempt from submitting the five case studies.